

SIMATS ENGINEERING ASSESSMENT TOOL1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5117

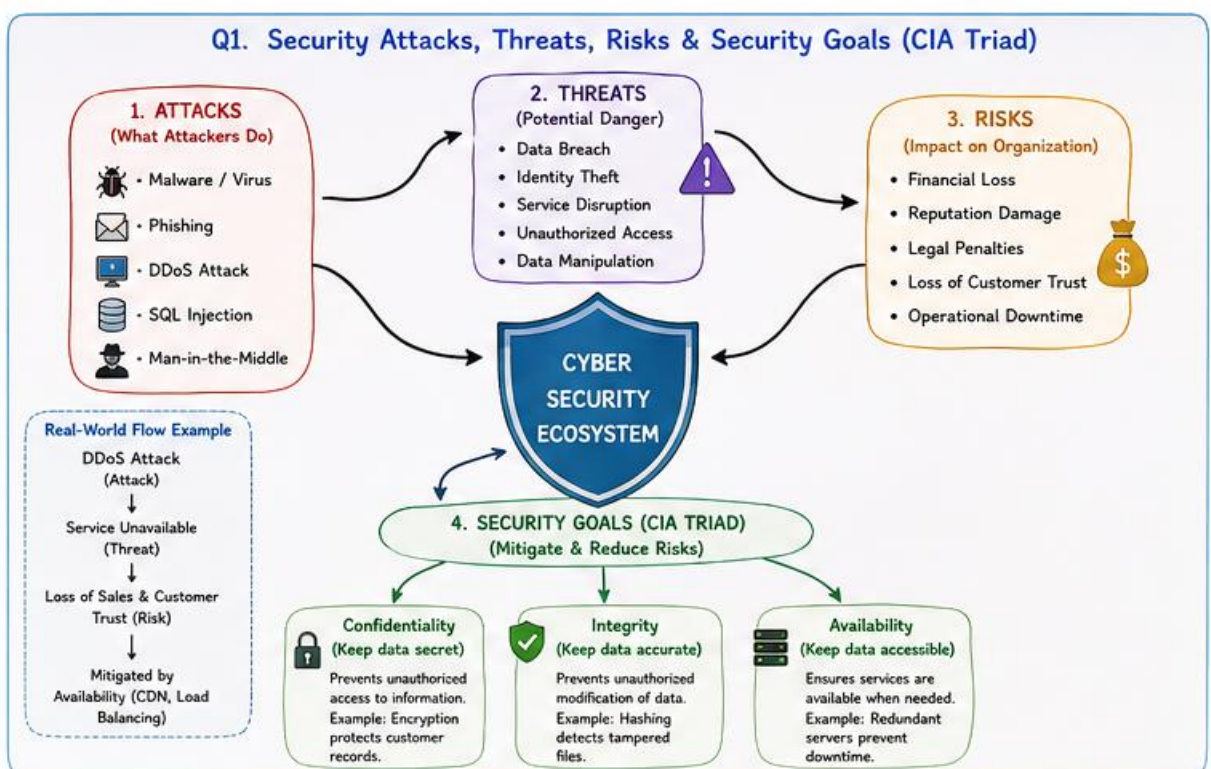
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CONCEPT MAPPING

Q1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:

- Show how each attack leads to specific threats and risks
- Map how security goals help mitigate them Provide one real-world example for each link.

ANS:



Concept Map Overview:

In cybersecurity, a security attack creates a threat that can lead to a risk for an organization. Attacks such as phishing, malware, and DDoS can cause data theft, data damage, or service disruption. To reduce these risks, security goals based on the CIA Triad are applied. Confidentiality protects sensitive information, Integrity maintains accurate data, and Availability ensures systems and services remain accessible.

Real-World Example:

Link: A botnet floods an e-commerce website's servers with malicious traffic during a holiday sale (**Attack**), crashing the servers and causing a total service outage (**Threat**), which results in millions of dollars in lost hourly revenue (**Risk**).

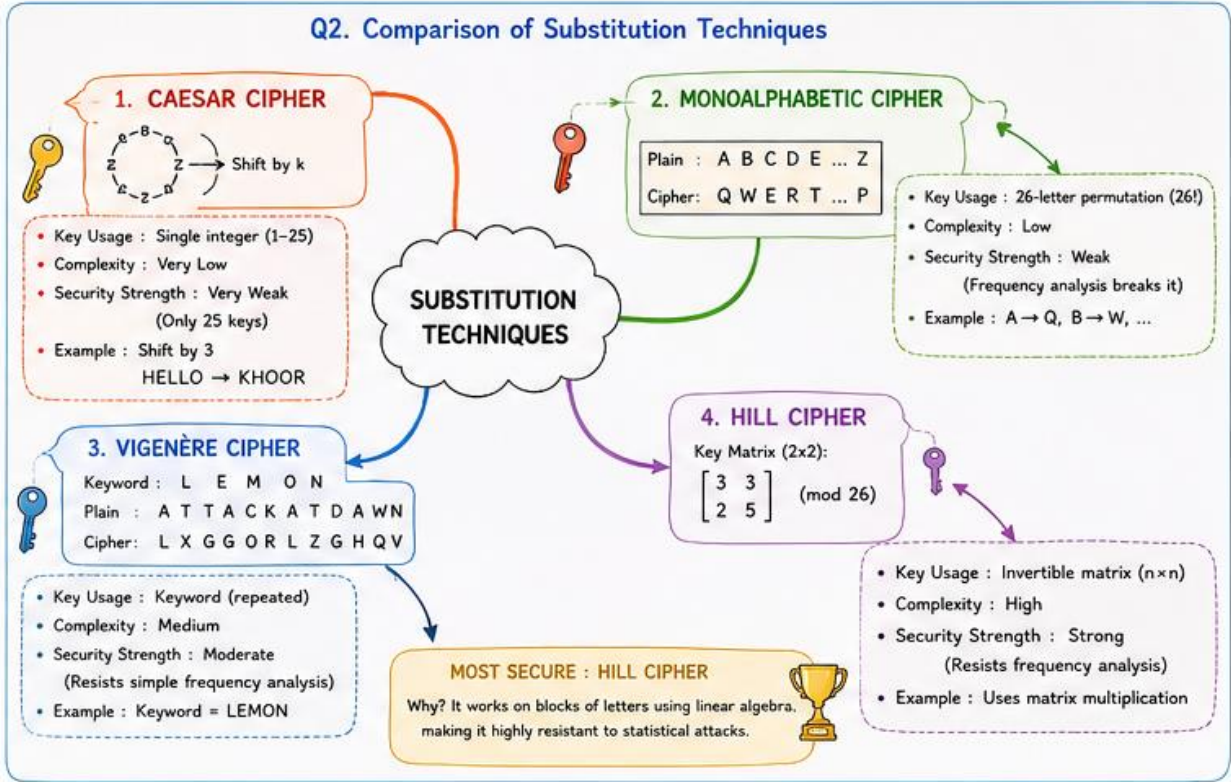
Mitigation: Ensuring **Availability** through geographically distributed Content Delivery Networks (CDNs) and auto-scaling cloud infrastructure absorbs the malicious traffic, keeping the storefront online for legitimate customers.

CIA Triad Mitigation Map:

The CIA Triad helps prevent cybersecurity risks by applying different protection methods. Confidentiality is maintained through encryption and access control to stop unauthorized data access. Integrity is protected using hashing, digital signatures, and backups to prevent data modification. Availability is ensured through backup systems, cloud infrastructure, and load balancing to keep services running during attacks. These security goals help organizations protect their data and maintain reliable operations.

Q2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

ANS:



1. Concept Map: Comparison of Substitution Techniques

This mapping categorizes the four classical substitution ciphers based on their structural properties, establishing a clear progression from simple single-character shifts to complex mathematical transformations.

Cipher Technique	Key Usage	Complexity	Security Strength
Caesar Cipher	A single integer (1-25) representing a fixed alphabetical shift.	Very Low. Simple addition/subtraction mechanism.	Very Weak. Trivial to break via brute-force (only 25 possible keys).

Monoalphabetic	A randomly shuffled 26-letter alphabet (key space of $26!$).	Low. 1-to-1 character substitution mapping.	Weak. Defeated easily by linguistic frequency analysis and pattern recognition.
Cipher Technique	Key Usage	Complexity	Security Strength
Vigenère Cipher	A text keyword repeated to match the length of the plaintext.	Medium. Polyalphabetic; shifts change dynamically per character.	Moderate. Masks standard letter frequencies but vulnerable to Kasiski examination.
Hill Cipher	An invertible $n \times n$ matrix (where n is the block size of characters).	High. Polygraphic; requires matrix multiplication and modular arithmetic.	Strong (Relatively). Highly resistant to ciphertext-only frequency analysis.

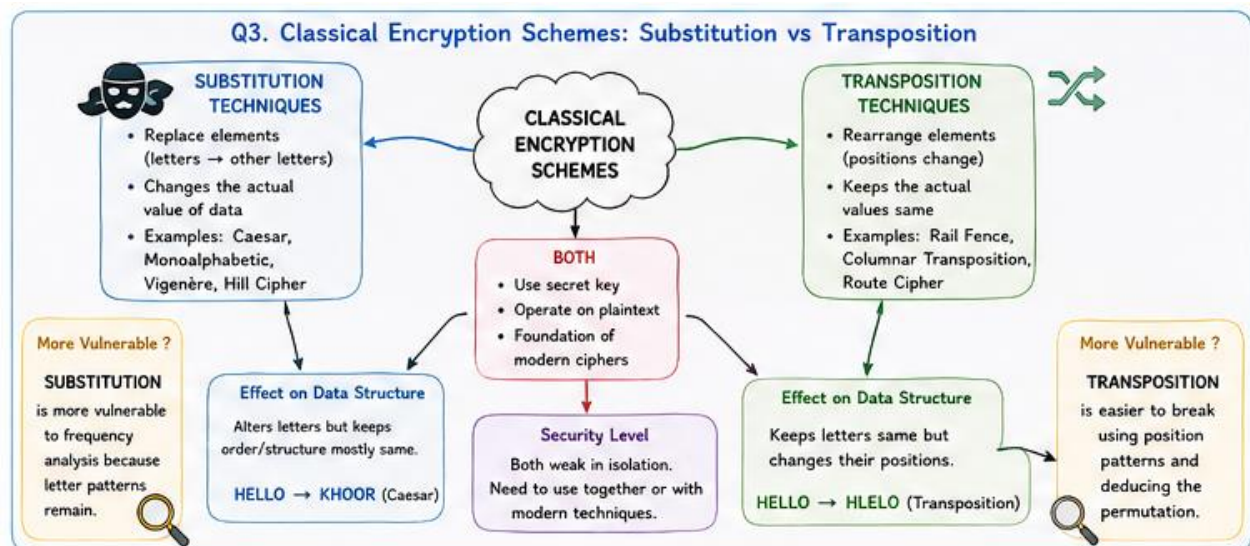
Q3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:

a. Show differences and similarities

i. Map how each affects the Data structure and Security level

ii. Analyse which is more vulnerable to attacks and why.

ANS:



1. Concept Map: Substitution vs. Transposition (Similarities & Differences):

Classical encryption schemes are mainly divided into Substitution Techniques and Transposition Techniques. Substitution techniques replace original characters with different characters while keeping the data order the same. Examples include Caesar Cipher, Monoalphabetic Cipher, and Vigenère Cipher. Transposition techniques rearrange the position of characters without changing the actual characters. Substitution provides better confusion, while transposition provides better diffusion. Both techniques improve data security, but classical methods have limitations. Substitution techniques are generally more vulnerable to frequency analysis attacks because repeated letter patterns remain, while transposition techniques can be attacked by analyzing character positions.

Substitution Techniques (e.g., Caesar, Monoalphabetic, Vigenère)

Data Structure Impact: Alters the value of the data at each specific index. The original sequence and spatial structure (like word lengths or repeating character positions) often remain entirely intact, but the characters themselves are transformed.

Security Level: Low to Moderate. While polyalphabetic substitution (like Vigenère) provides a moderate security level by masking single-letter frequencies, monoalphabetic substitution is extremely weak as it preserves the exact statistical footprint of the underlying language.

Transposition Techniques (e.g., Rail Fence, Columnar Transposition)

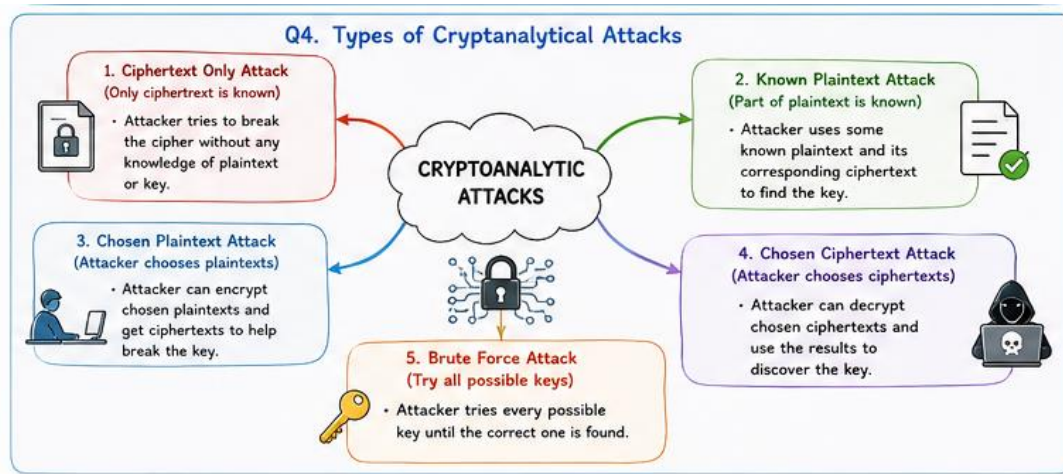
Data Structure Impact: Alters the index/location of the data while leaving the values untouched. The spatial relationships (words, consecutive letters) are shattered, scattering them across the data block.

Security Level: Low. Because it only moves letters around, an attacker possesses the exact unencrypted characters of the message. Security relies entirely on the complexity of the routing/rearrangement algorithm.

Q4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:

- Show relationships between these concepts
- Analyze how each contributes to secure communication
- Include one real-world application for each

ANS:



1. Concept Map: Relationships

Steganography is a technique used to hide secret information inside another medium such as images, audio, or video. The main concepts of steganography are **Data Hiding, Watermarking, and Survivability**. **Data Hiding** focuses on secretly embedding information to prevent unauthorized access. **Watermarking** adds hidden information such as ownership details or copyrights to protect digital content. **Survivability** ensures that the hidden information remains protected even after compression, editing, or transmission. Together, these concepts improve secure communication by keeping information confidential and protecting digital assets.

Data Hiding (The Method) $\rightarrow$ enables **Watermarking** (The Application) $\rightarrow$ which requires **Survivability** (The Defense).

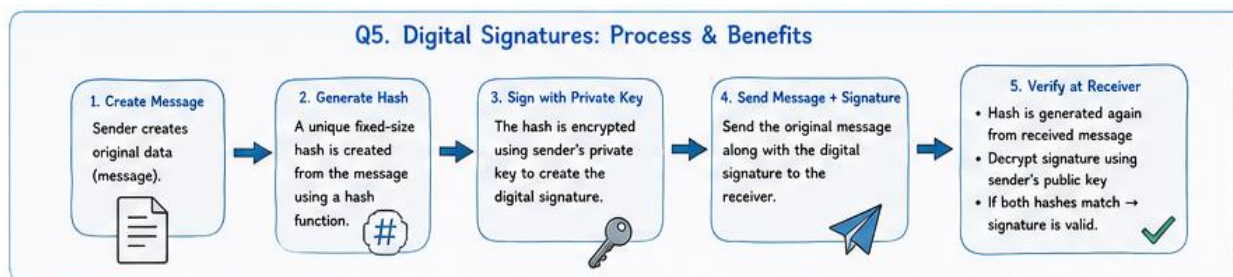
Real-World Example:

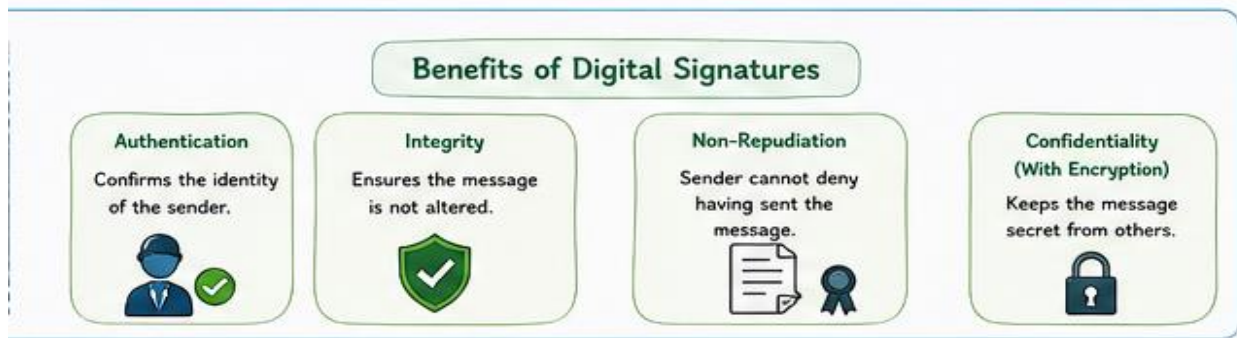
Data hiding is used in secure communication where a secret message is embedded inside an image and shared without revealing the message. Watermarking is used by photographers, filmmakers, and companies to add copyright information to their digital content and prevent unauthorized use. Survivability is used in digital platforms where hidden watermarks must remain detectable even after image resizing, compression, or editing. For example, online streaming services use digital watermarking to protect movies from illegal copying.

Q5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:

- Show how each concept is mathematically connected
- Map their role in encryption/decryption
- Analyze which concept is most critical for public key cryptography and justify.

ANS:





Concept Map: Mathematical Connections:

Number theory plays an important role in cryptography by providing mathematical techniques for secure encryption and decryption. Euler's Phi Function calculates the number of values relatively prime to a number and is used in key generation. Euler's Theorem provides the mathematical foundation for encryption and decryption operations. Fermat's Little Theorem is used for calculations involving prime numbers and modular arithmetic. The Extended Euclid Algorithm helps find the Modular Inverse, which is required to generate private and public keys. Among these concepts, the Modular Inverse and Euler's Phi Function are most critical for public key cryptography, especially in RSA, because they help create secure encryption and decryption keys.

Real-World Example:

In RSA encryption, two large prime numbers are selected to generate a public key and private key. Euler's Phi Function is used to calculate the value needed for key generation. The Extended Euclid Algorithm finds the modular inverse, which helps create the private key. During encryption, the public key is used to convert plaintext into ciphertext, and the private key decrypts the message back into original data. This mathematical process protects online banking, secure websites, and digital communication.

CIA Triad Mitigation Map:

Number theory concepts support the CIA Triad by providing strong cryptographic protection. **Confidentiality** is achieved through encryption algorithms that use modular arithmetic and secure key generation to prevent unauthorized access. **Integrity** is maintained through cryptographic operations that verify data authenticity and prevent modification. **Availability** is supported by efficient mathematical algorithms that allow secure communication systems to operate reliably. Among all concepts, **Modular Inverse is highly critical in public key cryptography because it enables the creation of private keys required for secure decryption in RSA systems.**